CS301: Deep Learning & Neural Computation

Lecture 04: Fundamentals of Deep Feedforward Networks

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Term: Fall Semester

Outline of Today's Lecture:
1. Motivation: Linear Classifiers vs Multi-Layer Perceptrons (MLP)
2. Activation Functions: Sigmoid, Tanh, ReLU, and GELU
3. The Backpropagation Algorithm and Chain Rule of Calculus
4. Vanishing and Exploding Gradient Dilemmas
5. Weight Initialization Schemes (He vs Xavier/Glorot)

Lecture 04 - Slide 2: Non-Linear Activation Functions

Why do we need non-linear activations?
Without non-linearities, any composition of linear layers collapses into a single linear map:
$y = W_2 * (W_1 * x + b_1) + b_2 = (W_2 * W_1) * x + (W_2 * b_1 + b_2) = W' * x + b'$

Key Activation Functions:
1. Sigmoid: $\sigma(z) = 1 / (1 + \exp(-z))$
   - Squeezes output between (0, 1)
   - Derivative: $\sigma'(z) = \sigma(z) * (1 - \sigma(z))$
   - Major Disadvantage: Vanishing gradient for $|z| >> 0$, not zero-centered.

2. Rectified Linear Unit (ReLU): $f(z) = \max(0, z)$
   - Constant gradient of 1 for $z > 0$ (combats vanishing gradient)
   - Computational efficiency: simple thresholding
   - Disadvantage: 'Dying ReLU' problem when neurons never activate (gradient = 0).

3. Leaky ReLU: $f(z) = \max(\alpha * z, z)$ where α is typically 0.01

Lecture 04 - Slide 3: Backpropagation and Weight Updates

Backpropagation computes the gradient of the scalar loss L with respect to all network weights W.

Mathematical Foundation - The Chain Rule:
dL/dW_ij = (dL/dz_j) * (dz_j/dW_ij) = delta_j * a_i
where delta_j is the local error term at layer l, and a_i is the activation from layer l-1.

Optimization with Stochastic Gradient Descent (SGD):
W_t+1 = W_t - eta * nabla_W L(W_t)
where eta is the learning rate.

Momentum and Adam Optimizer:
- Momentum accelerates SGD in directions of consistent curvature: v_t = gamma * v_t-1 + eta * nabla_
- Adam (Adaptive Moment Estimation) computes adaptive learning rates for each parameter using 1st a

Lecture 04 - Slide 4: Key Takeaways & Exam Preparation

Crucial Points for Midterm Review:
1. Universal Approximation Theorem: A feedforward network with a single hidden layer and non-linear a
2. Vanishing gradients predominantly occur in deep networks when sigmoid or tanh derivatives (< 0.25)
3. He Initialization initializes weights with variance Var(W) = 2/n_in, optimized specifically for ReLU acti
4. Practice Exercise: Show that if f(x) is linear, an L-layer deep network has no greater expressive capa